Database Systems Course Project

Full Database Design and Web Application Implementation

1. Project Overview

In this project, you will design, implement, and deploy a complete database-driven system. The goal is to model a real-world problem, implement it using a relational database, and develop a basic web application using PHP as the backend to interact with the database.

This project simulates the full database development lifecycle including conceptual design, logical design, implementation, query development, and web integration.

2. System Design and Requirements

- Students must design an original system that is realistic and sufficiently complex.
- The system must require at least 15 entities (relations).
- The system must clearly define business rules and user interactions.

Examples:

- Hospital Management System
- Online Learning Platform
- Event Management System
- E-commerce Marketplace
- Library Management System
- Hotel Reservation System
- Car Rental System
- University Registration System
- Fitness Club Management System

3. Conceptual Design – Entity Relationship Diagram (ERD)

- At least 15 entities (strong and/or weak entities).
- Primary keys clearly indicated.
- All relationships with correct cardinality.
- Participation constraints clearly defined.
- Weak entities where applicable.
- Multivalued attributes if applicable.
- Specialization/generalization if applicable.

4. Database Implementation

- Use MySQL.
- Write SQL scripts to create the database and tables.
- Define all constraints (PK, FK, UNIQUE, etc.).

- Populate the database with meaningful and consistent data.
- Insert at least 10 rows per table.

5. Advanced SQL Queries

- Create 15 complex SQL queries.
- Each query must use multiple tables (JOIN required).
- Include INNER JOIN and LEFT/RIGHT JOIN.
- Include GROUP BY and HAVING.
- Use aggregate functions (COUNT, SUM, AVG, MAX, MIN).
- 2 queries must include subqueries.
- 1 query must include 3 or more joined tables.
- Use ORDER BY and DISTINCT where appropriate.

For each query, include:

- The SQL statement.
- A short explanation of what the query does.
- A screenshot of the output.

6. Web Application Development (PHP Backend)

- Basic backend implementation using PHP.
- Use MySQL as the database.
- Use HTML for frontend interface.
- Implement proper database connection using mysqli.
- Implement CRUD operations for at least 3 major tables.
- Integrate at least 5 complex SQL queries into the web application.
- Provide basic form validation.
- Include a simple navigation menu and organized layout.
- Include sessions to track state between the pages.

7. Report Structure Requirements

- Cover page (Course name, Project title, Student names, IDs, Instructor, Date).
- System description and objectives.
- Entity Relationship Diagram (ERD).
- SQL implementation scripts.
- Sample data description.
- 15 SQL queries with explanations and outputs.
- Web application screenshots.